

# Reputation is Amortized Verification:

Inspection Games for Agent Economies

Paper 3 of the Harbor research program — B2/R7, executed

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## Abstract

A marketplace where AI judges grade AI work is an inspection game. We give three results. *Stage:* a bonded judge is deterred iff  $\rho dB \geq G$ ; the critical audit rate is  $\rho^* = G/(dB)$  under the keep-gain convention and  $\rho_c^* = G/(d(G+B))$  under confiscation — the settlement convention alone moves the headline audit budget by roughly 17%. *Tower:* when level  $k+1$  audits level- $k$  auditors sampled *sealed* from a pool spanning  $C$  disjoint cliques, rational bribery is all-or-nothing and profitable only while corrupt value exceeds  $CB$ ; below that threshold corruption decays geometrically at  $(1-\rho d)$  per level, so *finite* bond capital certifies a tower deep enough to drive corrupt value below any fixed unit — a depth logarithmic in that value. *Amortization:* as a judge's verified history grows, incentive-compatible audit schedules decline — lifetime audit spend falls from  $\Theta(T)$  (flat) to  $\Theta(\log T)$  (losses only if audited) to  $O(1)$ , with the constant  $aG^2/(2dBv)$  (cheats also surface at an independent rate  $r$ ). Reputation, on this account, is not a soft social layer bolted onto verification: it is the mechanism by which verification cost is amortized over a history. The results convert the source volume's central open conjecture (III.11.1) into a parameterized theorem with a named implementation, and every number reproduces from a bundled script.

*Express lane: deterrence holds iff audit-probability  $\times$  detection  $\times$  bond  $\geq$  corrupt gain; stacking auditors sampled sealed from  $C$  independent cliques shuts off the bribery-profitable phase, leaving a geometric collapse that finite total bond certifies at logarithmic depth; and history-indexed audit schedules make lifetime verification spend  $\Theta(\log T)$  or  $O(1)$  instead of  $\Theta(T)$  — formal statements in the boxes of §4, §5, §6.*

## 1 Introduction

**The scene.** A marketplace where AI agents do work and AI judges grade it. A judge posts a bond that is slashed if it is caught cheating. A judge can take a bribe: pass shoddy work, pocket  $G$ . So you audit the judges — but auditors can be bribed too, and now it looks like watchers all the way up, each level dearer than the last. Meanwhile the platform's accountant asks a different question: a judge with ten thousand verified grades is still being audited at the same rate as a newcomer — is that spend buying anything?

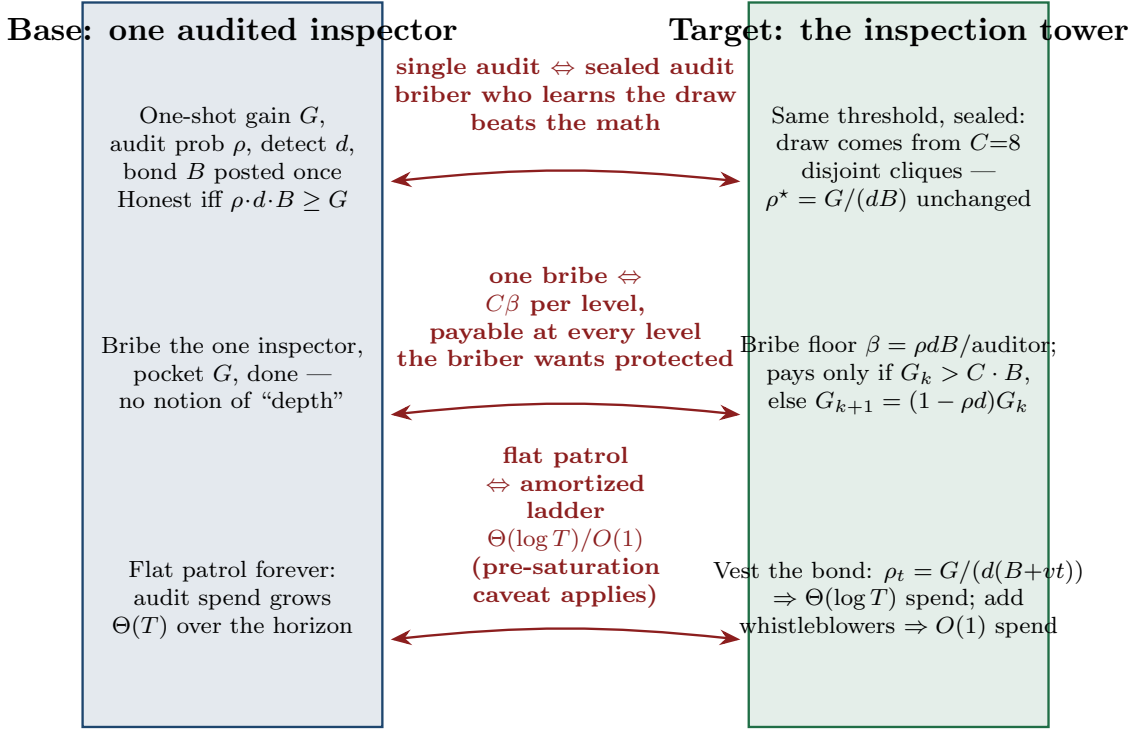
**The idea in one breath.**

*Honesty holds when expected punishment beats the bribe; the tower of auditors converges on finite bond capital because each level has geometrically less corruption left to protect, and drawing judges sealed from  $C$  independent camps buys down the depth by shutting off the phase where bribery still pays; and the audit rate an old judge needs falls with the reputation it would forfeit — so reputation is amortized verification.*

**The idea, before the formalism.** Traffic enforcement does not post an officer on every corner; it makes expected fine  $\times$  odds of being caught exceed the minutes saved by speeding.

That is the stage game: gain  $G$ , audit probability  $\rho$ , detection probability  $d$ , slashable bond  $B$ . The tower adds jury selection from disjoint pools: if each auditing layer is drawn sealed from  $C$  rival benches, a briber must pay all  $C$  benches to be safe — worthwhile only for enormous corruption, which then bleeds value paying for its own protection. And the amortization result is the actuarial half of the same logic: a judge with a long verified history has a large future income stream at stake, which is itself a punishment the platform did not have to fund — so the platform can audit it less, and the incentive constraint stays exactly binding. Figure 1 draws the correspondence. *Misread to preempt:* the tower result does *not* require infinitely many honest auditors, and the amortization result does *not* assume old judges are intrinsically nicer. The former needs only finite bonds and a sealed draw, because what each level protects shrinks geometrically once bribery stops paying; clique diversity ( $C > 1$ ) buys depth and capital, not convergence. The latter needs only that accumulated reputation is genuinely forfeit on capture — deterrence is re-derived, not assumed, at every  $t$ .

### R7 — reputation is amortized verification



*The tower prices depth, not detection: finite bond, sealed against the draw, certifies a depth logarithmic in the corrupt value — 27 levels at  $C=8$ , 53 at  $C=1$ .*

Figure 1: The relation map: the stage game (left) and the tower (right) are the same deterrence arithmetic, applied once and applied recursively. The arrows carry the load — a single committed audit maps to a *sealed* draw from  $C$  benches; one bribe maps to a bribe bill of  $C\beta$  that must be paid *at every level* the briber wants protected; and a flat patrol budget maps to a history-indexed schedule. The relation preserved across all three rows is *expected forfeiture  $\geq$  corrupt gain*; what changes is only who pays it and over what horizon. What  $C$  buys is depth and capital (53 levels and 2650 at  $C=1$  against 27 and 1350 at  $C=8$ ), not the difference between a tower that terminates and one that does not.

**Contributions.** (i) The stage-game threshold with its convention fork made explicit (§4) — the same game yields  $\rho^* = 0.25$  or  $\rho_c^* = 0.2083$  at our running parameters depending on whether capture confiscates the gain, and quoting one without pinning the settlement rule is a

reproducible way to be 17% wrong. (ii) The tower-contraction theorem (§5): sealed sampling from  $C$  disjoint cliques makes bribery uneconomical below corrupt value  $C B$ , after which corruption decays geometrically — so finite bond capital certifies an audit depth logarithmic in the corrupt value, with sealing identified as load-bearing rather than hygienic and the clique multiplier priced honestly (it removes the linear bribery phase; it is not what makes the tower converge). (iii) The amortization theorems (§6): flat auditing spends  $\Theta(T)$ ; the loss-only-if-audited schedule spends  $\Theta(\log T)$ ; independent revelation at rate  $r$  gives  $O(1)$  lifetime spend, with the constant  $aG^2/(2dBrv)$  — and which of the two holds is an empirical fork, not a modeling taste. (iv) The connection to the program’s repeated-game mechanization (§7): the discount-factor thresholds  $\delta \geq 1/3$  (grim) and  $\delta^* = 0.3425080314$  (graduated trigger), now CI-checked by Z3 and TLC/Apalache in the repository’s `proofs/economics/` directory, are the same deterrence arithmetic with continuation value in place of bonds. (v) The conversion of the source volume’s Conjecture III.11.1 into a parameterized theorem with a named implementation (§8). Every number is tagged [verified] (externally recomputable) or [internal] (regenerates from the bundled script `b2_tower.py`).

## 2 Related work, and what is actually new

The stage game is classical. Deterrence-by-expected-penalty is Becker [2]; the leader–follower and simultaneous inspection formulations, including the committed-inspector solution we use, are surveyed by Avenhaus, von Stengel and Zamir [1], whose treatment of inspector commitment and inspectee indifference we follow. Reputation-as-equilibrium-behavior is Kreps–Wilson [3] and Fudenberg–Levine [4]: a long-lived player’s continuation value disciplines short-run temptation. Batched verification of many items with few tests goes back to Dorfman’s group-screening design [5], which is the right lineage for pooled audit sampling, though we do not use pooling here.

*New, honestly stated:* the amortized-verification framing — deriving the *declining audit schedule* that keeps the incentive constraint exactly binding as history accumulates, and costing it end-to-end ( $\Theta(T)$  vs.  $\Theta(\log T)$  vs.  $O(1)$ , with the closed form  $aG^2/(2dBrv)$ ); the contraction theorem for a *stackable* audit tower whose auditors are themselves corruptible LLM judges, with model heterogeneity identified as the mechanism that supplies the disjoint cliques the contraction needs; and the observation that sealing the sampling draw is load-bearing (§5). *Imported:* everything in the previous paragraph. The contribution is closing a conjecture the source volume posed, with a named implementation and a measurement obligation — not a new solution concept.

## 3 The vocabulary, defined

*Express lane: skip to §4 — nothing in this section is needed to follow the theorems, only to read them comfortably; the two terms that are hypotheses rather than vocabulary, sealed and clique, are defined where they first do work, at the head of §5.* This paper is economics, and its terms are cheap to define but expensive to guess at. A software reader who has never taken a mechanism-design course loses nothing but the dictionary; here it is, in the order the terms are needed.

An *inspection game* is a two-player game between someone who may cheat and someone who may check, where checking costs something and catching pays something. Its defining feature — and the reason it is not simply a matter of “audit more” — is that it typically has no equilibrium in which either player behaves predictably: if the inspector always checks, cheating never pays and the checks are wasted; if the inspector never checks, cheating always pays. The stable outcome is that both randomise, which is why every audit rate in this paper is a probability rather than a schedule.

A *mixed strategy* is exactly that randomisation — a probability distribution over what to do rather than a choice. Two players are in a *mixed equilibrium* when each is *indifferent* between its options given what the other is doing, since a player who strictly preferred one option would simply take it. Solving for indifference is how the critical rates below are derived, and it is the whole technique.

A player’s behaviour is *incentive-compatible* when honesty is at least as profitable as the alternative — not when it is enforced or morally preferred. Everything this paper deters, it deters by making cheating cost more than it pays.

A *bond* is capital the judge posts before working and forfeits if caught cheating. It is the paper’s central instrument, and it differs from a wage in a way that matters: a wage is paid for output and withheld for non-performance automatically, whereas a bond is forfeited only when a *detection event* occurs. That gap — between misbehaving and being caught misbehaving — is what the audit probability  $\rho$  and detection probability  $d$  measure, and it is why  $\rho dB$  rather than  $B$  is the quantity a rational judge weighs. To *slash* a bond is to confiscate it on detection; the term is borrowed from proof-of-stake systems, where it means the same thing.

The two *settlement conventions* of §4 are the two natural answers to what happens to the corrupt gain itself when a cheat is caught. Under *keep-gain* the cheater keeps  $G$  and loses the bond; under *confiscation* the cheater loses both. They are not a modelling detail — they move the required audit budget by roughly 17% at the worked parameters, which is the paper’s first practical finding.

A *bribe floor* is the least a briber must offer a judge to make corruption worth its while: exactly the judge’s expected forfeiture  $\beta = \rho dB$ . Below that, no rational judge accepts.

Finally, *amortization* is the accounting sense of the word, not the algorithmic one: spreading a fixed cost over many uses so the per-use share falls. The paper’s title claim is that a reputation *is* this operation applied to verification cost — the reason a long-trusted judge can be audited less is that its history has already paid for the confidence.

## 4 The stage game and the convention fork

A judge grades one outcome. Cheating yields a one-shot gain  $G$ . The platform audits with probability  $\rho$  at cost  $a$  per audit; an audit detects a cheat with probability  $d$ ; detection slashes the judge’s posted bond  $B$ . Under the *keep-gain* convention the caught cheat retains  $G$  and forfeits  $B$ ; under the *confiscation* convention capture forfeits the gain as well.

**Theorem 1** (Stage deterrence). *A judge with one-shot corrupt gain  $G$ , facing audit probability  $\rho$ , detection probability  $d$ , and slashable bond  $B$ , is deterred iff  $\rho dB \geq G$ ; the critical committed audit rate is*

$$\rho^* = \frac{G}{dB} \quad (\text{keep-gain; Becker's condition}).$$

*If capture also confiscates the gain, the cheat payoff is  $G - \rho d(G+B)$  and the critical rate falls to*

$$\rho_c^* = \frac{G}{d(G+B)} < \rho^*.$$

*In the simultaneous (uncommitted) game, with  $a$  the audit cost and  $L$  the auditor’s loss from an uncaught cheat, the mixed Nash equilibrium is  $\rho^* = G/(dB)$ ,  $q^* = a/(dL)$ : the judge’s indifference pins the audit rate, the auditor’s indifference pins the cheat rate.*

*Why the same  $\rho^*$  twice.* The formula is unchanged across the two game forms not by coincidence but because both solve the same judge-indifference equation: commitment picks the cheapest rate that deters, and simultaneous play lands an equilibrium mix exactly there.

*Proof.* Keep-gain: the cheat payoff is  $G - \rho dB$ , nonpositive iff  $\rho \geq G/(dB)$ . Confiscation: replace  $B$  by  $G + B$  in the forfeiture. Mixed equilibrium: the judge mixes only if indifferent,  $G - \rho dB = 0$ ; the auditor mixes only if indifferent between auditing ( $-a - qL(1 - d)$ ) and not ( $-qL$ ), which solves to  $q^* = a/(dL)$ . Direct best-response checks confirm both indifferences numerically (§9).  $\square$

**Numbers by hand.** At the program’s running parameters  $G=10$ ,  $d=0.8$ ,  $B=50$ :  $\rho^* = 10/(0.8 \cdot 50) = 0.25$  [verified] — audit one grade in four, and the cheat payoff is exactly  $+0.000000$  at  $\rho^*$  [internal, `b2_tower.py`]. Under confiscation,  $\rho_c^* = 10/(0.8 \cdot 60) = 0.2083$  [verified]. The convention moves the required audit budget from 0.25 to 0.2083 — a 16.7% cut for free, *if and only if* your settlement rail actually claws back the gain. With  $a=1$ ,  $L=40$ :  $q^* = 1/(0.8 \cdot 40) = 0.03125$  [verified]. Figure 2 plots the frontier  $\rho dB = G$  in the  $(B, \rho)$  plane with this point marked. *Now you try:* double the bond to  $B=100$  — what is  $\rho^*$ ? (0.125: a bigger bond lets you audit half as often; capital and vigilance are substitutes along  $\rho dB = G$ .)

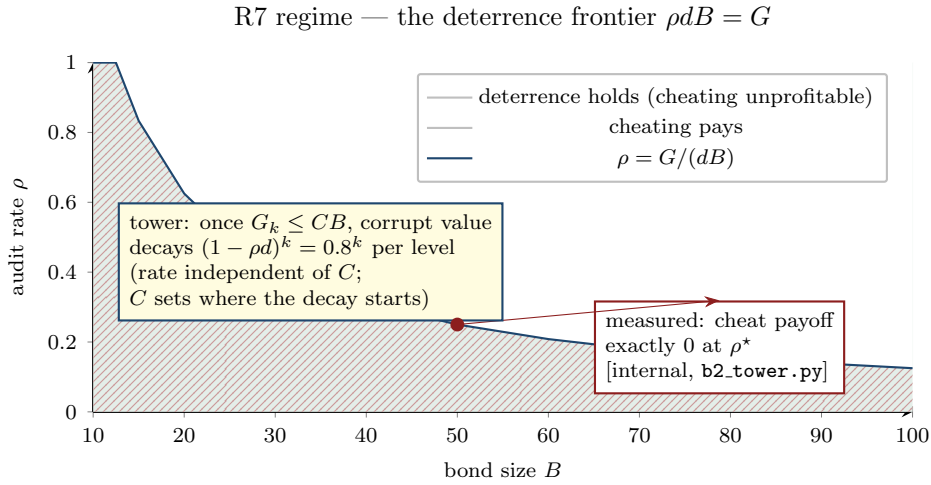


Figure 2: The regime diagram: the deterrence frontier  $\rho dB = G$  in the (bond  $B$ , audit rate  $\rho$ ) plane at  $G=10$ ,  $d=0.8$ . Above the curve  $\rho = G/(dB)$ , cheating is unprofitable; below it, cheating pays. The marked point is the running example  $(B, \rho^*) = (50, 0.25)$ , where the measured cheat payoff is exactly zero [internal, `b2_tower.py`]. Below the curve, the region is hatched rather than tinted so the two half-planes stay distinguishable in greyscale. The inset records the tower consequence of §5: below the bribery threshold  $G_k \leq CB$ , surviving corrupt value decays as  $(1 - \rho d)^k$  — a  $\times 0.8$ -per-level collapse at these parameters, at every  $C$ . What  $C$  controls is not the rate but where the decay begins.

## 5 The tower: bribing sealed juries from $C$ cliques

Two terms first, because both are hypotheses of the theorem below rather than hygiene recommendations. *Sealed* sampling means the briber must commit its money before learning which judge will handle the case; the entire clique multiplier vanishes without it, as the sealing discussion after the box shows. A *clique* means a pool of judges whose errors and loyalties are statistically and economically independent of another pool’s — distinct vendors running distinct model families. The word is borrowed from social structure, not from graph theory, and it carries no adjacency-matrix meaning in this paper.

Who audits the auditors? Let level  $k+1$  audit level- $k$  auditors. An auditor at any level is itself a bonded judge, so Theorem 1 prices its honesty: the minimum bribe an auditor rationally accepts is its expected forfeiture  $\beta = \rho dB$ . The platform’s one structural lever is *where auditors come from*: each audit is sampled, sealed, from a pool spanning  $C$  disjoint cliques — distinct

principals, distinct model families — such that a corrupt agreement inside one clique does not reach the others. A briber protecting corrupt value  $G_k$  at level  $k$  faces a choice: bribe, or absorb the audit.

**Theorem 2** (Tower contraction). *Let level  $k+1$  audit level- $k$  auditors, each audit sampled sealed (the draw unknown to the briber) from a pool spanning  $C$  disjoint cliques, each auditor bonded at  $B$  with audit parameters  $(\rho, d)$  and bribe floor  $\beta = \rho dB$ . Then the briber's expected net value is affine in the number of cliques bought,  $c \in \{0, \dots, C\}$ : buying  $c$  still leaves the un-bought  $(C-c)/C$  share of the sampling weight exposed to the same per-draw audit-and-detect risk  $\rho d$ , so the payoff has no interior optimum and rational bribery is all-or-nothing — partial bribery is (weakly) dominated by one of the two corners. Bribing all  $C$  is profitable iff*

$$G_k \rho d > C\beta \iff G_k > CB.$$

*Below that threshold bribery stops and corrupt value decays geometrically:*

$$G_{k+1} = (1 - \rho d) G_k.$$

*Consequently finite bond capital certifies a tower deep enough to drive surviving corrupt value below any fixed unit, at  $B$  per level. If  $G_0 \leq CB$  the decay is geometric throughout and the certified depth is  $\lceil \log G_0 / \log \frac{1}{1-\rho d} \rceil$ . If  $G_0 > CB$ , a linear phase of  $\lceil (G_0 - CB)/(C\rho dB) \rceil$  levels precedes it — during which the briber bleeds  $C\beta$  per level — followed by  $\lceil \log(CB) / \log \frac{1}{1-\rho d} \rceil$  geometric levels. Either way the depth is finite: logarithmic in the corrupt value that remains once bribery has stopped paying, plus that linear prefix if it was still paying at level 0. Raising  $C$  shortens or deletes the linear phase; it does not change the geometric rate  $(1 - \rho d)$ , which is what makes the tower converge.*

*Proof sketch.* Buying  $c$  of the  $C$  cliques costs  $c\beta$  up front (the seal forces payment before the draw is known) and retains value  $G_k$  if the draw lands on a bought clique (probability  $c/C$ ) or  $(1 - \rho d)G_k$  if it lands on an honest one (probability  $(C-c)/C$ ). Expected net value is

$$V(c) = G_k \left(1 - \frac{C-c}{C} \rho d\right) - c\beta = V(0) + c \left(\frac{G_k \rho d}{C} - \beta\right),$$

affine in  $c$ , so the maximum over  $c \in \{0, \dots, C\}$  sits at a corner: bribery is all-or-nothing. Bribing beats not bribing iff  $V(C) > V(0)$ , i.e.  $G_k \rho d > C\beta = C\rho dB$ , i.e. iff  $G_k > CB$ ; the audit bite  $\rho d$  cancels, leaving a pure capital comparison between the value protected and the bonds that must be compensated. Once  $G_k \leq CB$  the geometric recursion takes over, and the level count to drive the surviving value below one unit from there is  $\lceil \log \min(G_0, CB) / \log \frac{1}{1-\rho d} \rceil$ , each level requiring one bond  $B$ ; if  $G_0 > CB$ , the linear phase adds  $\lceil (G_0 - CB)/(C\rho dB) \rceil$  levels ahead of it.  $\square$

**Why sealing is load-bearing, not hygienic.** The factor  $C$  in the profitability threshold exists *only because the briber cannot see the draw*. If the sampling draw leaks — the briber learns *which* auditor will handle its case before the bribe is placed — it need buy exactly one auditor at price  $\beta$ , and bribery is profitable iff  $G_k \rho d > \beta$ , i.e. iff  $G_k > B$ : the clique multiplier vanishes and heterogeneity buys nothing. A leaked draw converts an eight-bench jury system into a single corruptible clerk. Sealed randomized selection (in deployment: VRF-selected honeypot audits, §8) is therefore part of the theorem's hypothesis, and auditing the *sampler* is part of the security boundary.

**Numbers by hand.**  $G_0 = 400$ , parameters as above,  $\rho = \rho^* = 0.25$ , so  $\beta = \rho dB = 10$  and  $\rho d = 0.2$ . At  $C=1$ : the profitability threshold is  $G_k > 50$ , so bribery still pays at first, and corruption bleeds only the bribe bill  $C\beta = 10$  per level —  $400 \rightarrow 390 \rightarrow 380 \rightarrow \dots$  — for 35 levels, until  $G_k$  reaches  $50 = CB$  and the geometric phase takes over for  $\lceil \log 50 / \log 1.25 \rceil = 18$  more:  $53$  levels,  $53 \times 50 = 2650$  of bond capital [verified, arithmetic; regenerates from

`b2_tower.py`]. At  $C=8$ : the threshold is  $G_k > 400$ , not met even at level 0, so there is no linear phase at all and the collapse is geometric from the start at  $(1 - \rho d) = 0.8$  per level:  $400 \rightarrow 320 \rightarrow 256 \rightarrow 204.8 \rightarrow \dots$ , reaching below one unit in  $\lceil \log 400 / \log 1.25 \rceil = 27$  levels, so total bond capital  $27 \times 50 = 1350$  [verified, arithmetic].

**What the clique multiplier actually buys.** Set the two runs side by side and the multiplier is precisely a factor of two in certified depth and capital — 53 levels and 2650 against 27 and 1350 — not the difference between a tower that converges and one that does not. Both converge. Raising  $C$  *removes the linear phase*, because the profitability threshold  $CB$  rises past  $G_0$ ; the geometric phase, and the rate  $(1 - \rho d)$  that drives it, are common to both. That is a real and budgetable saving, and it is the whole of the claim. *Now you try:* at  $C=2$ , what is the per-level bleed while bribery remains profitable, and where does the linear phase stop? ( $C\beta = 20$  per level:  $400 \rightarrow 380 \rightarrow 360 \rightarrow \dots$ , until  $G_k \leq CB = 100$  after 15 levels, where the decay turns geometric for a further 21 — 36 levels, 1800 of bond.)

**Why the per-level condition suffices.**  $V(c)$  scores a *single* level's deviation, so  $G_k > CB$  is an iff for one-level bribery, not for a bribery strategy. The distinction matters because buying level  $k+1$  protects  $G_k$  only from that level's audit: level  $k+2$  still threatens it, so a briber committed to the programme pays  $C\beta$  at every level for as long as the tower runs. Scoring the strategy rather than the level makes bribery *less* attractive, not more, so the contraction is unaffected — at  $C=1$  the all-bribe path spends  $35 \times 10 = 350$  to shepherd  $G_0 = 400$  down to 50 and then loses the 50 geometrically anyway, burning 87.5% of the protected value to arrive where it would have arrived regardless [verified, arithmetic]. The per-level condition is therefore the weaker and the safer of the two to state: it is what the affine argument establishes, and any multi-level bribery programme is dominated by it a fortiori.

## 6 Reputation is amortized verification

The stage game prices one grade. A platform judges  $T$  of them per judge lifetime, and flat auditing at  $\rho^*$  costs  $a\rho^*T = \Theta(T)$ . But an old judge is not a newcomer: it has a verified history of length  $t$ , and with per-period reputation rent  $v$  its accumulated stake  $vt$  — the future income its record commands — is forfeit on capture. That stake is punishment capital the platform did not fund, and an incentive-compatible audit schedule may spend it.

**Theorem 3** (Amortization). *Let a judge at history length  $t$  have reputation-at-stake  $vt$ , and let the audit schedule  $\rho_t$  be chosen so the one-shot deviation is weakly unprofitable at every  $t$ . Then:*

1. (**Flat.**) *The history-blind schedule  $\rho_t \equiv \rho^* = G/(dB)$  is incentive compatible and spends  $a\rho^*T = \Theta(T)$ .*
2. (**Model A: loss only if audited.**) *If a cheat is penalized only when an audit catches it, and capture forfeits both bond and stake, the binding schedule is  $\rho_t = G/(d(B + vt))$ , and lifetime spend is  $a \sum_{t < T} \rho_t = \frac{aG}{dv} \ln(1 + \frac{vT}{B}) + O(1) = \Theta(\log T)$ .*
3. (**Model B: independent revelation.**) *If cheats additionally surface without audits at rate  $r$  per period (whistle-blowers, downstream failures, tombstoned provenance), destroying the stake  $vt$  when they do, the binding schedule is  $\rho_t = \max(0, \frac{G - rvt}{dB})$ , which reaches zero at  $t^* = G/(rv)$ ; lifetime spend is bounded by the closed form*

$$a \sum_{t \geq 0} \rho_t \longrightarrow \frac{aGt^*}{2dB} = \frac{aG^2}{2dBrv} = O(1).$$

*Under each schedule the deviation value is  $\leq 0$  pointwise at every  $t$ : reputation does not*

*weaken deterrence, it funds it.*

*Proof.* (2) The deviation value at  $t$  is  $G - \rho_t d(B + vt)$ ; setting it to zero gives the schedule, and the spend is a harmonic-type sum:  $a \sum_t G/(d(B + vt)) \approx \frac{aG}{dv} \ln(1 + vT/B)$ . (3) The deviation value is  $G - \rho_t dB - rvt$  (audit slash plus the expected independent loss); the binding schedule is linear and hits zero at  $t^* = G/(rv)$ , so the spend is the area of a triangle with base  $t^*$  and height  $a\rho^* = aG/(dB)$ :  $\frac{1}{2} \cdot \frac{G}{rv} \cdot \frac{aG}{dB} = \frac{aG^2}{2dBrv}$ , independent of  $T$ .  $\square$

**Numbers by hand — with the caveat a referee would otherwise find.** Parameters  $G=10$ ,  $d=0.8$ ,  $B=50$ ,  $a=1$ ,  $v=0.6$ ,  $r=0.05$ , horizon  $T=200$ . Flat spend:  $1 \cdot 0.25 \cdot 200 = 50.00$  [verified]. Model A: 25.58 measured against closed form  $\frac{aG}{dv} \ln(1 + vT/B) = 25.50$  [internal, `b2_tower.py` / verified]. Model B: measured spend 35.08 against the  $O(1)$  limit  $aG^2/(2dBrv) = 41.67$  [internal/verified]. **Note that  $35.08 \neq 41.67$ , and the discrepancy is not error:** at these parameters  $t^* = G/(rv) = 333 > T = 200$ , so the audit schedule has not yet reached zero within the measured horizon — *the  $O(1)$  curve had not saturated at  $T=200$* . The measured 35.08 is the partial triangle; the closed form is its full area, attained only for  $T \geq t^* = 333$ . Run the horizon past  $t^*$  and the cumulative spend flattens and stays there. Figure 3 shows both halves of this: panel A is the measured range, where all three curves are near-linear and the  $O(1)$  curve is in fact the second-steepest; panel B runs the same three schedules to  $T=1500$ , where flat reaches 375.00, Model A 61.46 and is still climbing, and Model B stops dead at 41.79 from  $t^*$  onward and never spends again [verified, `paper3_amortization_figures.py`]. (The plateau is 41.79 rather than 41.67 because the closed form is the area of the continuous triangle while the schedule is summed period by period; the excess is exactly half the first term,  $\rho^*/2 = 0.125$ .) That is why the pre-asymptotic gap at  $T=200$  is a check on the closed form rather than a strike against it: read against  $t^*$ , not against the limit, 35.08 is precisely the partial area the theorem predicts. The incentive checks pass pointwise: maximum deviation value  $+1.78 \times 10^{-15}$  under both schedules — machine-epsilon zeros, i.e.  $\leq 0$  up to float epsilon, exactly the binding-constraint signature [internal]. *Now you try:* at what  $T$  do Model A and flat differ by a factor of 4? (Solve  $\frac{aG}{dv} \ln(1 + vT/B) = \frac{1}{4}a\rho^*T$ ; at these parameters  $T \approx 779$  — the log curve’s lead widens without bound.)

**The fork is empirical.** Whether a deployment lives in Model A or Model B is a question about the world, not the model: *do cheats surface without audits?* If bad grades eventually reveal themselves — failed downstream builds, provenance tombstones, user reports — then  $r > 0$  and lifetime verification spend per honest judge is a constant. If a cheat undetected at audit time is silent forever, only the  $\Theta(\log T)$  schedule is safe. The deployment telemetry that estimates  $r$  (the rate at which sanctioned outcomes were first flagged by non-audit channels) is therefore part of this paper’s measurement obligation, not an implementation detail.

## 7 The discount-factor connection: claim signaling, mechanized

The program’s repeated-game track prices the same deterrence with a different currency. In the corrected claim-signaling stage game — payoffs (3, 3) mutual-truth, (0, 4)/(4, 0) sucker/defector, (1, 1) mutual-claim — the one-shot deviation gain is  $g = 1$  and the per-round punishment loss is 2. By the one-shot deviation principle, a grim trigger is subgame-perfect — cooperation survives every history, not just the equilibrium path — iff  $1 \leq 2\delta/(1 - \delta)$ , i.e.  $\delta \geq 1/3$ ; the gentler three-round graduated trigger is subgame-perfect iff  $1 \leq 2(\delta + \delta^2 + \delta^3)$ , whose threshold is the unique real root in  $(0, 1)$  of  $2\delta^3 + 2\delta^2 + 2\delta - 1 = 0$ , numerically  $\delta^* = 0.3425080314$ . These are not folklore numbers: they are CI-mechanized in the repository’s `proofs/economics/` directory — `delta-threshold.z3` proves existence, uniqueness in  $(0, 1)$ , and location in  $[0.34, 0.35]$  of the root by real-algebraic decision procedure; `claim_signaling.tla` model-checks the discounted one-shot-deviation invariant under both TLC and Apalache; and `sweep-delta.sh` confirms the empirical crossover ( $\delta = 0.35$  on the integer grid) matches the closed form. Every claim in this paragraph is [verified] by that CI suite.

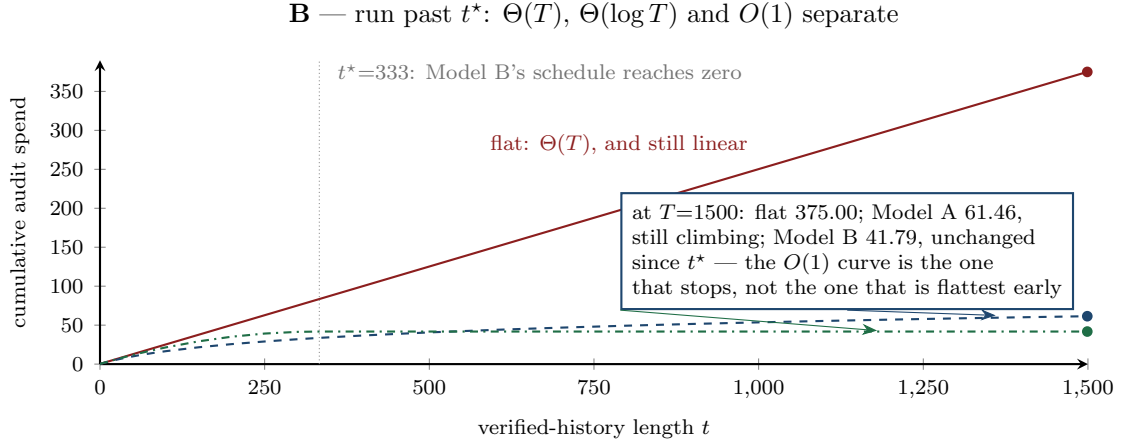
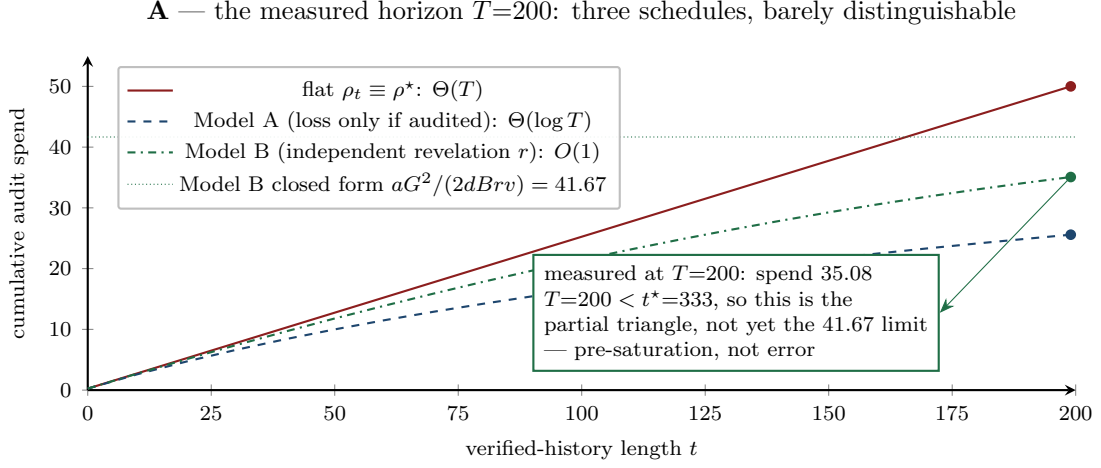


Figure 3: Cumulative audit spend against verified-history length  $t$  under the three incentive-compatible schedules of Theorem 3, at the running parameters. **A**: the horizon the paper measures,  $T=200$ . All three curves are near-linear here and Model B — the  $O(1)$  schedule — is the *second-steepest*: on this range the asymptotics are not visible, and the measured 35.08 sits below the closed-form 41.67 because saturation requires  $t^* = G/(rv) = 333$ . That gap is pre-asymptotic, not error. **B**: the same three schedules run to  $T=1500$ , past  $t^*$ . Now the growth classes separate on the page: flat reaches 375.00 and keeps climbing linearly, Model A reaches 61.46 and is still climbing (logarithmically), and Model B stops dead at 41.79 from  $t^*$  onward and never spends again. Model A's cumulative spend crosses *above* Model B's at  $t=532$ : past that point the  $O(1)$  schedule is permanently the cheaper of the two, which is the reversal panel A cannot show. The plateau 41.79 exceeds the continuous closed form 41.67 by  $0.125 = \rho^*/2$ , the expected discrete-sum overshoot on a linear ramp. Line styles differ (solid / dashed / dash-dot) so the three series survive greyscale printing [verified: `paper3_amortization_figures.py`, seed 20260816; cross-checks against `b2_tower.py`].

The connection to Theorem 3 is exact, not analogical. In Kreps–Wilson [3] and Fudenberg–Levine [4], deterrence is funded by *continuation value*: the discounted future surplus a deviator forfeits. In the inspection game it is funded by *bond plus stake*:  $d(B + vt)$  at audit,  $rvt$  by revelation. Model A’s schedule  $\rho_t = G/(d(B + vt))$  is precisely the statement that as endogenous continuation value  $vt$  accumulates, exogenous verification  $\rho_t$  can retreat while the total deterrent stays pinned at  $G$  — the bonded game continuously deforms into the repeated game. The discount thresholds  $\delta \geq 1/3$  and  $\delta^* = 0.3425$  are the limiting case  $B = 0$ ,  $\rho = 0$ : all deterrence carried by the future. Reputation is amortized verification in both formalisms; the inspection game just itemizes the invoice.

## 8 Conjecture III.11.1, parameterized

The source volume’s central open question — Conjecture III.11.1, the grading-oracle problem — asked whether a recursive rate-the-raters scheme can terminate. Two external reviews proposed answers that seemed to compete: one mandated a heterogeneous judge mix (distinct model families, to break correlated mode collapse) with VRF-selected honeypot tasks so a judge never knows when it is audited; the other derived the local condition  $\rho dB > G$  and insisted a recursive tower needs a per-level contraction plus a grounded termination. Neither had the theorem. Fused, they are it:

- *Model heterogeneity supplies the cliques.* The disjointness that Theorem 2’s  $C$  counts is statistical and economic independence of judge pools — exactly what distinct principals running distinct model families provide, and what a homogeneous pool (one vendor, one model,  $C=1$ ) does not. The cost of  $C=1$  is a linear bribery phase before the geometric one, roughly doubling certified depth and bond capital at the worked parameters (53 levels and 2650 against 27 and 1350); heterogeneity buys the removal of that phase, not the difference between a tower that converges and one that does not (§5).
- *VRF honeypots implement the sealed audit.* A verifiable random function [6] selecting honeypot tasks at rate  $\rho^*$  is precisely the sealed sampling the contraction hypothesis requires: the draw is unpredictable to the briber (sealing) yet publicly auditable after the fact (so the sampler itself cannot be quietly corrupted).
- *The termination is named, not wished.* The tower roots in an exogenously honest anchor — the operator at  $n=1$ , a bonded arbitration market at scale — and Theorem 2 prices the depth: 27 levels and bond capital 1350 at the running parameters [verified, arithmetic]. Concretely, this is what the judge ceremony of the reputation chapter is buying: the re-audit sampling rate is  $\rho^*$ , the “distinct architectural weights” quorum is the  $C$  in  $CB$ , the VRF injection is the seal, and the depth-and-capital line item ( $27 \times B$ ) is a number a platform can put in a budget rather than an assumption it has to hope holds.

The conjecture thereby becomes a *parameterized theorem* — deterrence at  $\rho^* = G/(dB)$  per level, contraction  $(1 - \rho d)$  under sealed  $C$ -clique sampling with the bribery cutoff  $G_k \leq CB$ , amortized lifetime cost by Theorem 3 — plus a *measurement obligation*: does the deployed judge pool actually satisfy sealed multi-clique sampling, and what are  $d$ ,  $r$ , and the cross-clique collusion correlation? Those are numbers to estimate from audit telemetry, not assumptions to wave at.

## 9 Reproducibility

Every number in this paper regenerates deterministically from one self-contained script, `skills/harbor-result` (numpy + matplotlib; program seed convention 20260816, though the computation is deterministic). Its output at the running parameters, verbatim:

```

rho* (keep-gain)      = G/(dB)      = 0.2500
rho* (confiscation)   = G/(d(G+B)) = 0.2083
q* (mixed NE)        = a/(dL)      = 0.0312
judge indifferent at rho*: cheat payoff = +0.000000
C=1: 400.0 390.0 380.0 370.0 360.0 350.0 340.0 330.0 320.0
C=8: 400.0 320.0 256.0 204.8 163.8 131.1 104.9 83.9 67.1
levels to drive corrupt value below 1 unit (geometric regime): 27;
total bond capital = B per level * levels = 1350
flat: 50.00 (Theta(T))
model A: 25.58 (Theta(log T); closed form 25.50)
model B: 35.08 (O(1); closed form  $aG^2/(2dBrv)$  = 41.67; hits 0 at  $t^*=333$ )
IC check A: max deviation value = +1.78e-15 (<=0)
IC check B: max deviation value = +1.78e-15 (<=0)

```

The script also verifies the mixed equilibrium by direct best-response check (auditor indifference  $-1.25 = -1.25$ ) and renders the three-panel summary figure. The claim-signaling thresholds of §7 regenerate independently via the CI suite in `proofs/economics/` (Z3, TLC, Apalache, and the  $\delta$ -sweep).

## 10 Honest boundaries

### What this paper does not say.

- *The headline is convention-dependent.*  $\rho^* = 0.25$  under keep-gain, 0.2083 under confiscation — a  $\sim 17\%$  swing from the settlement rule alone. Quote neither without stating which convention your slashing contract implements; a platform whose escrow claws back the gain but which budgets audits at  $G/(dB)$  is over-auditing, and one that budgets at  $G/(d(G+B))$  without clawback is under-deterring.
- *Sealed sampling is a hypothesis, not a garnish.* A briber who learns the audit draw before placing bribes faces threshold  $G_k > B$ , not  $G_k > CB$ : the clique math collapses to  $C=1$  (§5). The sampler — and its VRF, and whoever operates it — is inside the trusted computing base of Theorem 2.
- *The tower needs an honest root.* Contraction shrinks corruption level over level; something exogenously honest must sit at the top — the operator in a small deployment, a bonded arbitration market at scale. The theorem prices the depth to that root; it does not conjure the root.
- *$d$ ,  $r$ ,  $v$ , and cross-clique correlation are measured quantities, not assumptions.* Detection probability comes from audit outcomes; the revelation rate  $r$  from non-audit provenance of sanctions; cross-clique collusion correlation from judge-pair agreement statistics. “Disjoint cliques” is falsifiable and must be monitored: two nominally rival benches fine-tuned from the same base model may fail disjointness in exactly the correlated-error cases that matter.
- *The  $\Theta(\log T)$ -vs- $O(1)$  fork is empirical.* Model B’s constant lifetime spend holds only if cheats surface without audits ( $r > 0$ ). And at the session parameters the  $O(1)$  claim is demonstrated pre-asymptotically:  $t^* = 333 > T = 200$ , so the measured spend 35.08 sits below, not at, the 41.67 limit (§6).
- *One-shot gains, risk-neutral bribers, stationary parameters.*  $G$  is per-grade; an adversary who can correlate many small cheats below audit granularity, or who values capture asymmetrically, needs the repeated-game extension of §7 with the stake re-derived. The

machine-epsilon IC residuals ( $+1.78 \times 10^{-15}$ ) are zeros of a float computation, stated as “ $\leq 0$  up to float epsilon,” not as exact algebra.

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